

DEPARTMENT OF ELECTRICAL ENGINEERING

Electronic Devices & Circuits Laboratory (EDC LAB)

EXPERIMENT 1: V–I Characteristics of PN Junction & Zener Diodes, and Study of Clipper & Clamper Circuits

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Course: Electronic Devices & Circuits Lab
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1. Aim

1. To plot the forward- and reverse-bias V – I characteristics of a PN junction diode and determine its cut-in voltage (V_γ), static forward resistance (R_{DC}), dynamic forward resistance (r_d), and reverse saturation current (I_s).
2. To plot the forward- and reverse-bias V – I characteristics of a Zener diode (1N4733A) and determine its Zener breakdown voltage (V_Z) and Zener resistance (R_Z).
3. To design, simulate, and test positive, negative, and biased (two-level) diode clipper circuits and observe their clipping levels.
4. To design, simulate, and test positive and negative diode clamper circuits and measure the resulting DC voltage shift.

2. Apparatus and Software Tools Required

Table 1: Equipment and Components Specifications

S.No.	Component / Equipment	Specification	Quantity
1	PN Junction Diode	1N4007 / 1N4148	1
2	Zener Diode	1N4733A ($V_Z = 5.1$ V, 1 W)	1
3	Resistors	1 k Ω , 10 k Ω	As required
4	Capacitor	0.1 μ F	1
5	DC Voltage Sources	0–30 V Variable, +1 V, –1 V Reference	As required
6	Function Generator / AC Source	1 kHz Sine wave, 4 V _{pp}	1
7	Simulation Software	LTspice	–

3. Part A: PN Junction Diode Characteristics

3.1 Theoretical Background

A PN junction diode exhibits non-linear current-voltage characteristics described by the Shockley diode equation:

$$I = I_s \left(e^{\frac{V}{\eta V_T}} - 1 \right) \quad (1)$$

where I_s is the reverse saturation current, $V_T = \frac{kT}{q}$ is the thermal voltage (≈ 26 mV at room temperature), and η is the ideality factor.

- **Forward Bias:** When anode is positive relative to cathode, the barrier potential is reduced. Below the cut-in voltage ($V_\gamma \approx 0.6\text{--}0.7$ V for Silicon), current is negligible. Above V_γ , current increases exponentially.
- **Reverse Bias:** When cathode is positive relative to anode, the barrier height increases, allowing only a minute reverse saturation current I_s to flow until reverse breakdown occurs.

3.2 Circuit Diagrams and Simulation Waveforms

3.2.1 PN Junction Diode – Forward Bias

The forward bias circuit consists of a 1 k Ω current-limiting resistor and a PN diode. The DC voltage supply is swept from 0 V upward.

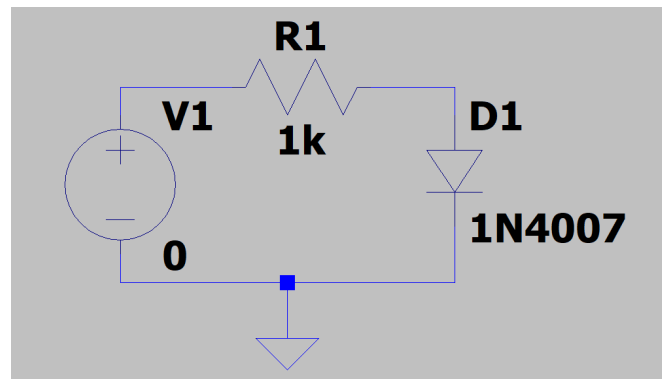
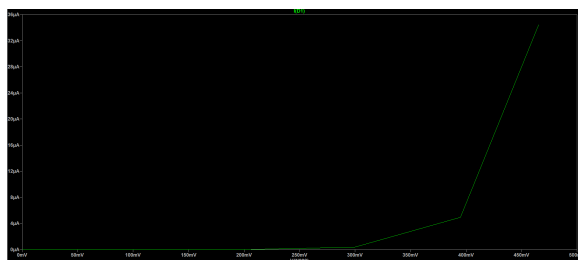
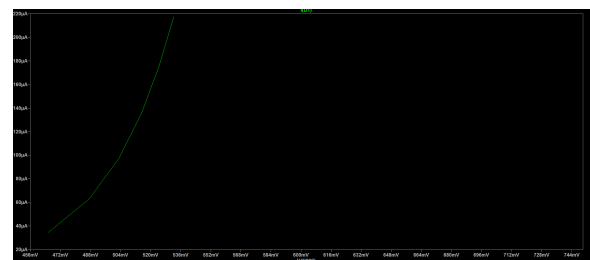


Figure 1: PN Junction Diode Forward Bias Circuit Schematic.



(a) Low Voltage Sweep (0 V to 0.5 V)



(b) Knee Region Sweep (0.5 V to 0.75 V)

Figure 2: PN Diode Forward V – I Characteristics Simulation Plots.

3.2.2 PN Junction Diode – Reverse Bias

The reverse bias circuit reverses the diode polarity to measure the reverse saturation current across a 0–30 V sweep.

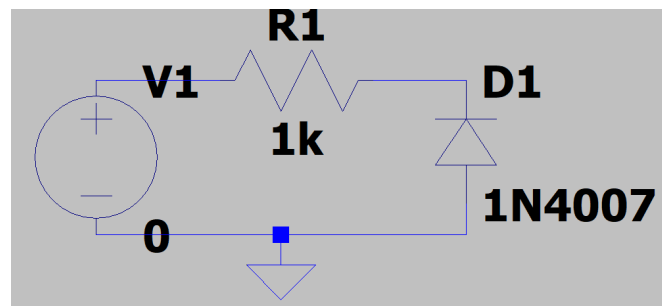


Figure 3: PN Junction Diode Reverse Bias Circuit Schematic.

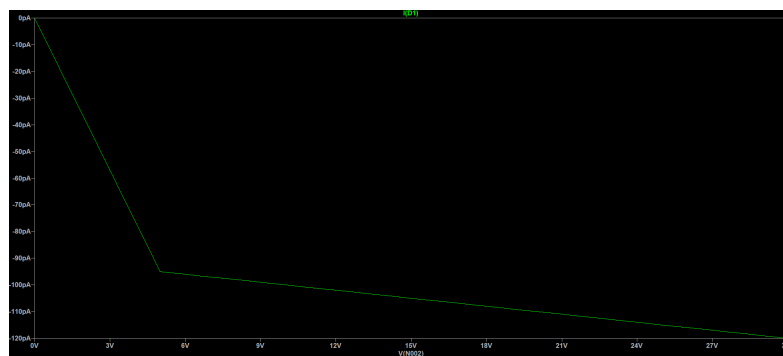


Figure 4: PN Diode Reverse Characteristic (0 V to 30 V).

4. Part B: Zener Diode Characteristics

4.1 Theoretical Background

A Zener diode is designed to operate safely in the reverse breakdown region. When reverse-biased beyond the Zener breakdown voltage (V_Z), the voltage across the diode remains virtually constant over a wide range of currents. In forward bias, it behaves like an ordinary PN junction diode.

4.2 Circuit Diagrams and Simulation Waveforms

4.2.1 Zener Diode – Forward Bias

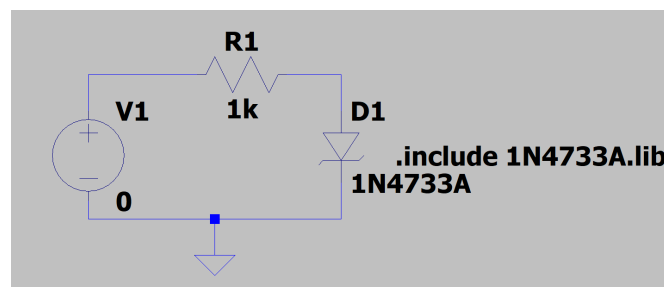
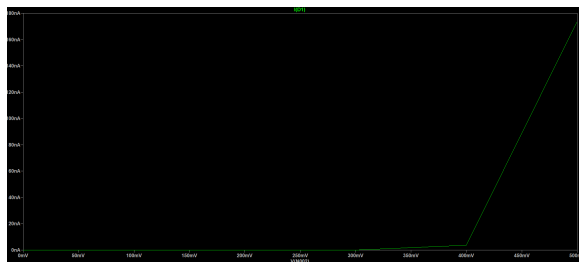
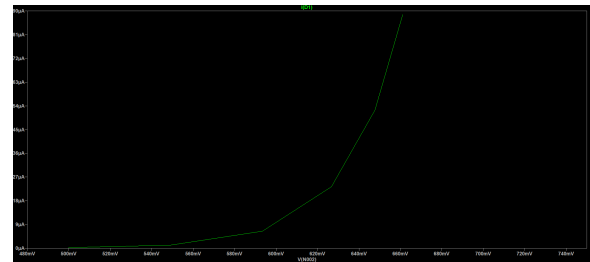


Figure 5: Zener Diode Forward Bias Circuit Schematic.



(a) Sub-knee Voltage Region (0 V to 0.5 V)



(b) Forward Conduction Region (0.5 V to 0.75 V)

Figure 6: Zener Diode Forward Bias Simulation Plots.

4.2.2 Zener Diode – Reverse Bias and Breakdown

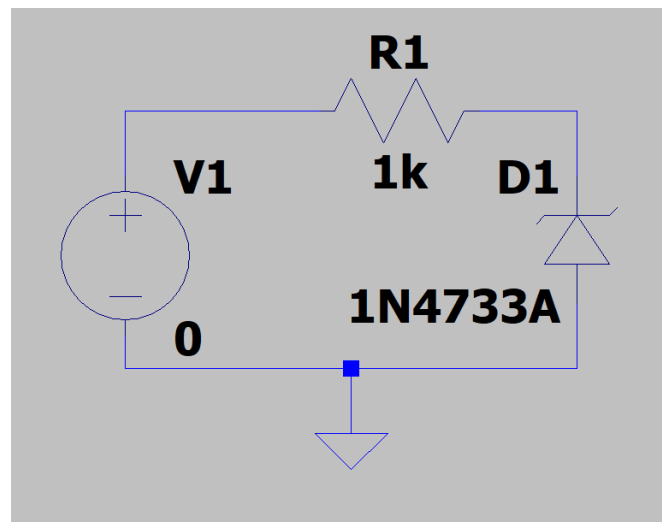


Figure 7: Zener Diode Reverse Bias Circuit Schematic.

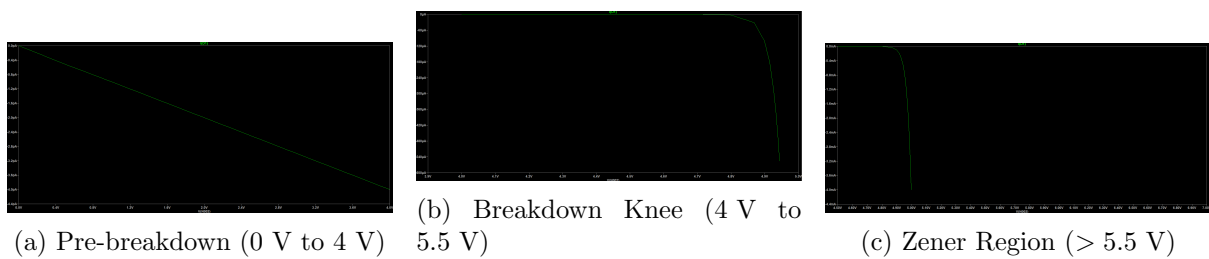


Figure 8: Zener Diode Reverse Breakdown Simulation Response.

5. Part C: Diode Clipper Circuits

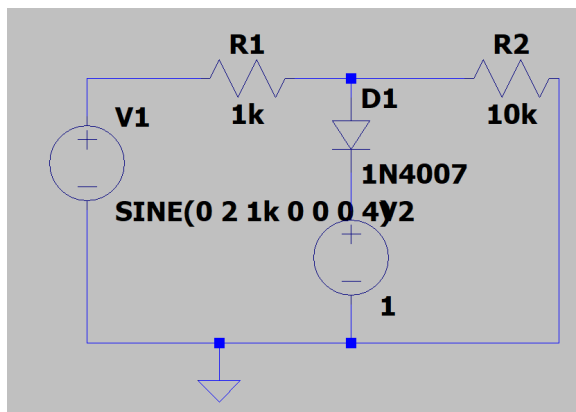
5.1 Theoretical Background

Clippers are wave-shaping circuits that remove portions of an applied signal wave above or below specified reference voltage levels without distorting the remaining part.

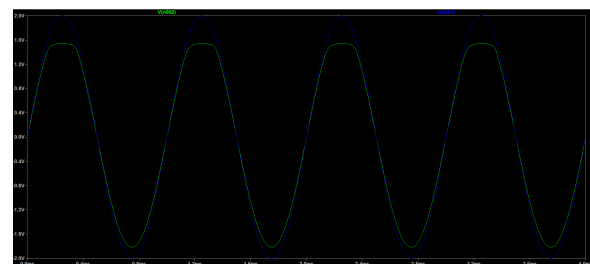
- **Positive Clipper:** Clips off the signal waveform above a threshold voltage ($V_{ref} + V_\gamma$).
- **Negative Clipper:** Clips off the signal waveform below a threshold voltage ($-V_{ref} - V_\gamma$).
- **Biased Clipper:** Combines positive and negative clippers to clamp peak amplitudes on both positive and negative half-cycles.

5.2 Circuit Diagrams and Simulation Waveforms

5.2.1 Positive Clipper (Clipping above +1 V reference)



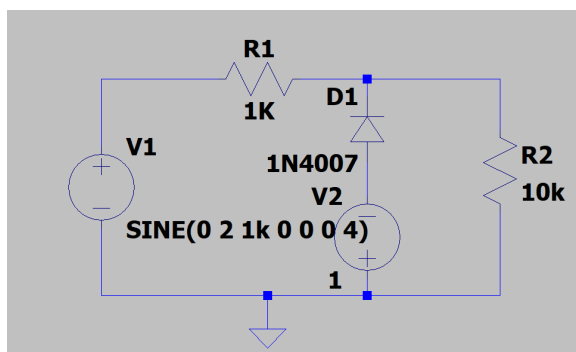
(a) Circuit Diagram



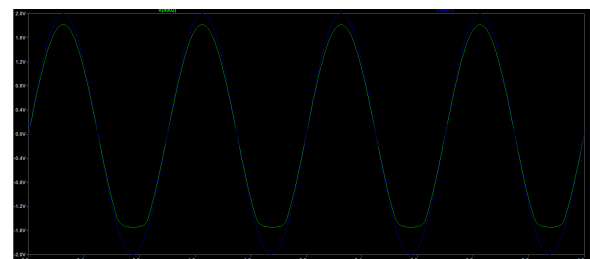
(b) Input and Clipped Output Waveforms

Figure 9: Positive Clipper Circuit and Transient Simulation.

5.2.2 Negative Clipper (Clipping below -1 V reference)



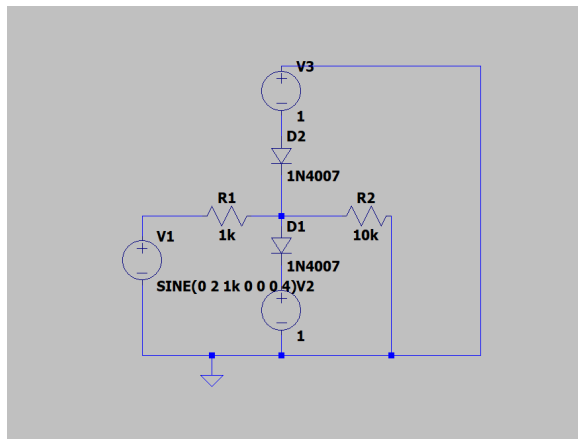
(a) Circuit Diagram



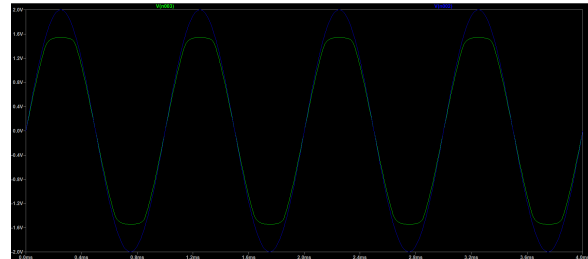
(b) Input and Clipped Output Waveforms

Figure 10: Negative Clipper Circuit and Transient Simulation.

5.2.3 Biased Clipper (Two-Level Clipper)



(a) Circuit Diagram



(b) Input and Two-Level Clipped Waveforms

Figure 11: Biased Clipper Circuit and Transient Simulation.

6. Part D: Diode Clamper Circuits

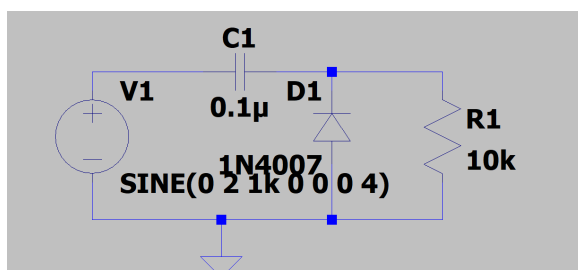
6.1 Theoretical Background

Clampers (DC restorers) shift a waveform vertically by adding a DC level to the input signal without altering the shape or peak-to-peak amplitude of the signal. They consist of a capacitor, diode, and load resistor.

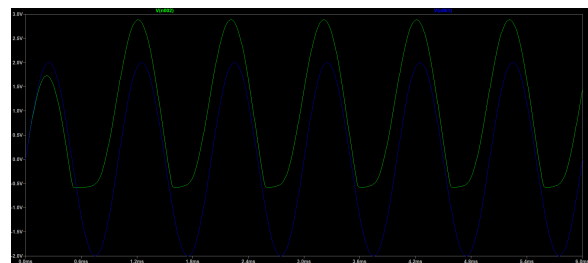
- **Positive Clamper:** Shifts the entire waveform upward so that its minimum peak is clamped near 0 V (or diode drop).
- **Negative Clamper:** Shifts the entire waveform downward so that its maximum peak is clamped near 0 V.

6.2 Circuit Diagrams and Simulation Waveforms

6.2.1 Positive Clamper



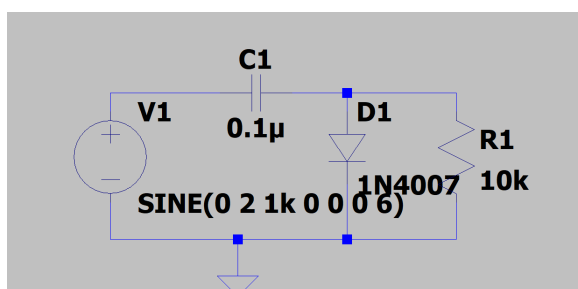
(a) Circuit Diagram



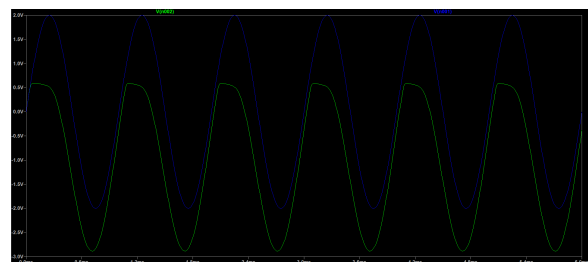
(b) Input and Upward Shifted Output Waveforms

Figure 12: Positive Clamper Circuit and Transient Simulation.

6.2.2 Negative Clamper



(a) Circuit Diagram



(b) Input and Downward Shifted Output Waveforms

Figure 13: Negative Clamper Circuit and Transient Simulation.

7. Observation Tables (Experimental Data Placeholder)

Note: Experimental readings to be populated post hardware testing.

7.1 Forward and Reverse Bias Observations

Table 2: PN Junction Diode and Zener Diode Conduction Data

PN Junction Diode (1N4007)			Zener Diode (1N4733A)		
S.No.	V_f (V)	I_f (mA)	S.No.	V_r (V)	I_r (mA)
1			1		
2			2		
3			3		
4			4		
5			5		
6			6		
7			7		
8			8		

7.2 Clipper and Clamper Summary Table

Table 3: Wave-shaping Circuits Performance Summary

Circuit	Input V_{pp} (V)	Theoretical Threshold/Shift	Measured Output Level
Positive Clipper	4.0	$+1.0\text{ V} + V_\gamma$	
Negative Clipper	4.0	$-1.0\text{ V} - V_\gamma$	
Biased Clipper	4.0	$+1.0\text{ V}$ and -1.0 V limits	
Positive Clamper	4.0	$+V_p - V_\gamma \approx +1.3\text{ V}$ shift	
Negative Clamper	4.0	$-V_p + V_\gamma \approx -1.3\text{ V}$ shift	

8. Conclusion

The simulation and theoretical study for Experiment 1 were successfully performed. The forward and reverse characteristics of both the standard PN diode and the Zener diode match theoretical expectations. Furthermore, the transient response of the positive, negative, and biased clipper circuits as well as the positive and negative clamper circuits demonstrate accurate signal shaping and DC level restoration.